

In vivo CRISPR screening identifies regulators of hyperplastic and hypertrophic adipose remodelling in zebrafish

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eLife Assessment

This is a well-written study that presents a **solid** genetic screen to identify regulators of adipose morphology and remodeling in zebrafish. The authors generated a rigorous screening platform based on live, whole animal imaging and statistical methods that revealed both novel and known genes critical for adipose regulation. This work is **valuable** because it provides several candidate genes relevant to metabolic health and a quantitative screening pipeline that will be beneficial for future studies. A limitation of the study is that it precludes a definitive distinction between developmental and remodeling effects.

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Abstract

Adipose tissues exhibit a remarkable capacity to expand, regress, and remodel in response to energy status. The cellular mechanisms underlying adipose remodelling are central to metabolic health. Hypertrophic remodelling - characterised by the enlargement of existing adipocytes - is associated with insulin resistance, type 2 diabetes, and cardiovascular disease. In contrast, hyperplastic remodelling - in which new adipocytes are generated - is linked to improved metabolic outcomes. Despite its clinical importance, the regulation of hypertrophic and hyperplastic adipose remodelling remains poorly understood. In this study, we first leveraged human genetic and transcriptomic data to identify candidate genes involved in adipose remodelling. We then developed a quantitative imaging pipeline to assess hyperplastic and hypertrophic morphology in zebrafish subcutaneous adipose tissue, and applied it in an F0 CRISPR mutagenesis screen targeting 25 candidate genes. This screen identified six genes that significantly altered adipose morphology; including Sushi Repeat Containing Protein (SrpX) - a gene with previously unknown roles in adipose. Among the identified genes, *foxp1b* mutants were notable for inducing hypertrophic morphology. To investigate further, we generated stable loss-of-function alleles for both zebrafish *foxp1* genes. We found that *foxp1b* mutants display a developmental bias towards hypertrophic adipose growth but fail to undergo further hypertrophic remodelling in response to a high-fat

diet - suggesting that early developmental patterning constrains later adaptability to diet. Together, these findings establish a scalable and tractable in vivo screening platform for identifying regulators of adipose remodelling, and reveal a potential developmental influence on the capacity for diet-induced adipose expansion.

Introduction

Adipose tissues exhibit a remarkable capacity to remodel and restructure themselves in response to the energetic status, the environment, physiological demands, or disease. Adipose remodelling involves alterations in both adipocyte number (addition or removal of adipocytes) and adipocyte size (expansion or reduction in adipocyte size). Increases in adipocyte size (hypertrophy) can trigger hypoxia, necrosis, altered lipid flux, inflammation and insulin resistance (1–3). Accordingly, a hypertrophic adipose morphology – adipose characterised by few, large adipocytes – is associated with increased cardiometabolic disease (4–6). Contrastingly, a hyperplastic adipose morphology – characterised by many, small adipocytes – is associated with more metabolically beneficial outcomes (7,8). Indeed, reductions in the hyperplastic potential of mouse adipose is associated with decreased insulin sensitivity (9), and disruption of adipocyte progenitor differentiation, proliferation and renewal leads to adipose hypertrophy and ensuing metabolic dysfunction (10). Strikingly, adipose hypertrophy is increased and preadipocyte frequency is reduced in diabetic obese patients (relative to non-diabetic obese) (11), and genetic predisposition for Type 2 diabetes, but not obesity, is associated with an impaired ability to recruit new adipose cells to store excess lipids in the subcutaneous adipose tissue (12). Despite its clinical relevance, the genetic, cell and molecular processes that determine the capacity for hyperplastic and/or hypertrophic adipose plasticity are largely unknown but will be key for treating and understanding adipose-related disease.

The zebrafish provides a tractable and powerful in vivo model for studying adipose growth and remodelling dynamics. Zebrafish adipose tissue is morphologically similar to mammalian white adipose tissue, comprising large adipocytes with a single dominant cytoplasmic lipid droplet (13,14). Transcriptomic profiling further supports this conservation, with RNA-Seq data revealing strong molecular similarity between zebrafish and mammalian white adipose tissue (15). Functionally, zebrafish adipose depots respond dynamically to nutritional status – expanding following a high-fat diet and regressing following food restriction (16,17). Further, prolonged high-fat or overfeeding has been shown to induce diabetic phenotypes in zebrafish (18,19), reinforcing the model's relevance to metabolic disease. One of the key advantages of the zebrafish model is its optical accessibility and amenability to fluorescent imaging. Lipophilic dyes such as Nile Red can be used to label lipid droplets (LDs) in vivo (13,20), and LD-associated protein reporters now allow real-time visualisation of lipid droplet dynamics in living animals enabling rapid assessment of adiposity and LD morphology (21,22). Additionally, transgenic tools are increasingly available to label and manipulate zebrafish adipocytes (23,24). Importantly, zebrafish are amenable to targeted CRISPR screens, and recent studies have utilised this methodology to examine roles for candidate obesity genes (25,26). However, to date, the zebrafish model has not been applied to systematically study the cellular growth properties of adipose tissue—specifically the regulation of hyperplastic versus hypertrophic remodelling.

Here, we leverage transcriptomic data from human subcutaneous adipose to identify genes associated with hyperplastic/hypertrophic adipose morphology. We then establish an image-based method for quantifying hyperplastic/hypertrophic morphology signatures in zebrafish, and apply it in a CRISPR screen targeting 25 candidate morphology genes. We identify six genes that alter adipose morphology, three whose disruption induces hypertrophic morphology (*foxp1b*, *txnipa* and *mmp14b*), and three that induce hyperplastic morphology (*ptenb*, *cxcl14* and *srpx*). To our knowledge, one of these – Sushi Repeat Containing Protein (*Srpx*) – has no previously characterised

roles in adipose biology. For functional follow-up, we focus on *Foxp1*, a transcription factor known to regulate stem and progenitor cells maintenance across multiple tissues. We generated new stable mutant lines for the two zebrafish *foxp1* genes (*foxp1a* and *foxp1b*), and find that mutation of *foxp1b* leads to a developmental bias towards hypertrophic growth, but an inability to undergo hypertrophic growth in response to high-fat diet. These findings suggest that early developmental patterning of adipose morphology may influence subsequent capacity to respond to diet-induced remodelling.

Methods

Identification of differentially expressed genes associated with small or large adipocytes in human subcutaneous adipose

Differentially expressed genes (DEGs) associated with large or small adipocytes in human SAT were obtained from Honecker et al., 2022. Briefly, 3,727 genes were significantly associated with adipocyte area by both categorical and continuous statistical models ($FDR < 0.05$) (27 [↗](#)). 2,190 DEGs were associated with small adipocyte area, 1,537 DEGs associated with large adipocyte area (Suppl. Table 1 [↗](#)). Expression within human WAT cell types for each of the candidate morphology genes was determined using data from (28 [↗](#)) via the Single Cell Portal (https://singlecell.broadinstitute.org/single_cell [↗](#)). Scaled mean expression for each of the morphology DEGs was assessed within each of the 16 previously defined WAT cell clusters (adipose stem and progenitor cells (ASPCs), mesothelium, pericyte, adipocyte, macrophage, endothelial, LEC, monocyte, T cells, dendritic cells, smooth muscle cells, b cells, mast cells, natural killer cells, endometrium, and neutrophils). Of the 3,727 morphology genes, the following were removed: 410 genes without annotations in bulk RNA-Seq dataset, 250 genes not found in the Single Cell Portal dataset, and 87 genes not expressed in the 16 WAT cell types. Hierarchical clustering of the remaining 2,980 morphology genes based on WAT cell type expression was performed in Morpheus (<https://software.broadinstitute.org/morpheus> [↗](#)) using an average linkage method and one minus Pearson correlation metric (Suppl. Fig. 1 [↗](#)). Morphology genes enriched in ASPCs were identified as either (i) belonging to the 75 morphology DEGs within the ASPC cluster, or (ii) one of the top 50 morphology DEGs with highest scaled mean expression in ASPCs (Suppl. Table 2 [↗](#)). Three genes were present in both categories (*CXCL14*, *KAZN* and *SPARC*), resulting in 122 ASPC-enriched morphology DEGs (Suppl. Table 2 [↗](#)). Ribosomal genes were excluded along with genes that did not code for proteins, resulting in 102 candidate morphology genes (Suppl. Table 2 [↗](#)).

Prioritisation and ranking of candidate adipose morphology genes

Enrichment analysis on the 102 candidate genes described above was performed using GProfiler (<https://biit.cs.ut.ee/gprofiler/gost> [↗](#)), with the 3,727 original genes as a background gene set. Six enriched GO biological process gene sets were identified, comprising; three ‘developmental’ terms (GO:0048869, GO:0009888 & GO:0032502), one ‘differentiation’ term (GO:0030154), and two more generic ‘process’ terms (GO:0048869 & GO:0048523) (Suppl. Table 3 [↗](#)). For each of the 102 morphology genes, beta coefficients between SAT expression levels and the 23 cardiometabolic traits from the METSIM study were clustered in Heatmapper (www.heatmapper.ca [↗](#)) using an average linkage clustering method and Pearson distance measurement method (Suppl. Fig. 2 [↗](#)). Correlations between SAT expression and 23 cardiometabolic traits from 770 males as part of the METSIM study were obtained from (29 [↗](#)). The following genes were not found in the METSIM data: *EIF4EBP3*, *PDF*, *SELENOM* and *SELENOS*. Candidate body mass index (BMI) and waist-to-hip ratio (WHR) genome-wide association study (GWAS) genes were identified by (i) downloading summary statistics for BMI and WHR from the GWAS Catalog (<https://www.ebi.ac.uk/gwas/> [↗](#)), (ii) identifying unique variants associated with the traits at genome-wide significance ($p < 5 \times 10^{-8}$), (iii) calculating proxy variants in high linkage disequilibrium in CEU population ($R^2 > 0.8$), (iv) calculating ‘LD blocks’ that encompass each lead and associated proxy variants and (v) identifying

genes that overlap with the LD blocks. The DIOPT Ortholog Finder (https://www.flyrnai.org/cgi-bin/DRSC_orthologs.pl) was used to identify zebrafish orthologs of the candidate human morphology genes based on the weighted score. The ‘alignment & scores’ function in DIOPT was used to calculate protein similarity between human and zebrafish genes. scRNA-Seq scaled mean expression and % of cells expressing candidate genes from Yang Loureiro et al. (2023) were obtained from the Single Cell Portal.

Zebrafish husbandry and maintenance

Zebrafish were housed in the Queen’s Medical Research Institute (QMRI) facility at the University of Edinburgh, UK. Standard husbandry was followed on a recirculating system and a 14h:10h light to dark cycle. Feeding regimen was as described in Tandon et al., 2025. The wild-type line used were WIKs maintained at the QMRI facility. Zebrafish experiments were conducted in accordance with the UK Animals (Scientific Procedures) Act 1986 under the project licence PP9112175.

F0 CRISPR screen in zebrafish

We followed general CRISPR screen methodology (30–32). Between three and five guide RNAs (gRNAs) were designed to target each gene using the UCSC Genome browser track ZebrafishGenomics (31) and CHOPCHOP software. gRNAs were selected based on off-target and efficiency scores, along with location in gene (targeting early exons, functional domains, and avoiding exons with lengths divisible by three). Pooled in vitro transcription of gRNAs was performed as per Wu et al (2018), and RNA was cleaned using Zymoclean columns (Zymo Research, Cat. # R1013). Prior to injection, gRNAs were incubated with Cas9 protein (New England Biolabs, Cat. # M0646T) and heated for 5 mins at 37°C. Mutagenesis of individual gRNAs was verified using T7E1 assays (New England Biolabs, Cat. # M0302S). gRNAs were injected into zebrafish embryos at the one-cell stage as per Wu et al 2018, along with a Fast Green dye to screen for successfully injected embryos. Identical injection mixes lacking the gRNAs (Cas9-only control) were used for control groups. 50 injected embryos were placed per petri dish and screened for survival at 5 days post fertilisation (dpf). At 5 dpf, injected larvae were placed in a small volume of system water in a 1L nursery tank without running water. From 10 dpf, a slow flow of system water was introduced. Larvae were raised at a density of 25 per 1L nursery tank, before being transferred to a 3L tank at 21 dpf. Survival was further assessed at 21 dpf. Feeding regimen was as described for standard husbandry. At 36–42 dpf, fish were euthanised and Nile Red staining performed as described previously (16).

Image-based adipose morphology profiling in zebrafish

For each Nile Red-stained juvenile zebrafish, we took two images on a Leica M205 FCA fluorescence stereomicroscope with a 1x objective: a whole animal image to allow use to measure total adiposity and standard length (SL), and a higher magnification image of the lateral subcutaneous adipose tissue (LSAT) along the zebrafish flank (hereafter referred to just as subcutaneous adipose or SAT). Images were always taken of the right side of the fish, as zebrafish adipose shows left-right asymmetry, with the right side containing the majority of pancreatic adipose. Images were first processed in Fiji/ImageJ and the ‘Clear Outside’ function was used to reduce image size and focus on the SAT area of interest or wider zebrafish. LD SAT images were then imported to Cellpose v2.2 and segmented using the ‘cyto’ model (33). Automated segmentations were assessed and manually adjusted to correct errors. Segmented masks and object outlines were imported to Fiji/ImageJ where xy coordinates for each segmented LD along with LD measurements were taken (including Feret’s diameter for LD size). ImageJ macros used for the importation of Cellpose masks and measurement of xy coordinates are available at github.com/jeminchin/morphology_screen. Total adipose area and SAT area were segmented using Ilastik software as previously described (34). Standard length was measured using the line tool in Fiji/ImageJ.

Statistical analysis of morphology values for the zebrafish F0 CRISPR screen

To calculate hyperplastic/hypertrophic adipose morphology we followed methodology from (35 [DOI](#)), and used an initial dataset of 194 Nile Red-stained wild-type zebrafish to fit a generalised additive model (GAM) to capture the relationship between SAT area and mean SAT LD diameter. Morphology value was calculated as the deviation of individual mean LD diameters from the fitted model. To compare hyperplastic/hypertrophic adipose morphologies between control and CRISPR mutants, we first trained a GAM from either (i) experiment-specific control data (eg, Cas9-only injected controls), or (ii) control data pooled from multiple experiments ($n = 343$ control zebrafish). These trained models were then applied to mutant test data to assess morphology values (i.e. mean mutant LD diameter deviation from the training model) using the *mgcv* package in R. To assess differences in morphology values between CRISPR mutants and controls, we conducted Kolmogorov–Smirnov (KS) tests to assess overall distribution shape differences. P values were corrected for multiple tests using Benjamini-Hochberg FDR correction (23 tests performed, adjusted p value < 0.05). We performed KS tests using both experiment-specific and pooled control data. Secondly, to complement the KS tests, we log-transformed SAT area and mean SAT LD diameter to linearise the relationship and performed linear mixed-model regression with replicate as a random effect to control for replicate-specific effects. Using this method, we assessed both intercept and slope differences between CRISPR mutant (pooled replicates) and experiment-specific controls or pooled controls. P values were corrected for multiple tests using Benjamini-Hochberg.

Generation of *foxp1a* and *foxp1b* mutant zebrafish lines

foxp1a and *foxp1b* zebrafish mutant lines were made using standard CRISPR-Cas9 methods (34 [DOI](#)). For both genes we targeted the highly conserved DNA-binding Forkhead domain (FHD). The location and sequence of the CRISPR gRNAs were: *foxp1a* – 5'-GACGAGTGGAGAATGTGAAG-3' targeting the FHD in exon 14, and *foxp1b* – 5'-GATAACGAAGCATACGTGAA-3' targeting the FHD in exon 15. We generated two alleles - ed116 is a 13bp indel in the forkhead domain of *foxp1a* resulting in a frameshift and premature stop codon, and ed125 is a 4bp indel in the forkhead domain of *foxp1b* resulting in a frameshift and premature stop. Primers to amplify locus surrounding lesions for T7E1 assays or sequencing were: ed116 forward – 5'-GCCAGATTGGACTGGATGTT-3', ed116 reverse – 5'-TTATTTCCAGGCCATTCTGG-3', ed125 forward – 5'-TTCAGTTTCAGCTCCTTCCTTC-3', ed125 reverse – 5'-TGGAAGTCAAGTACCAGCA-3'. To genotype ed116, primers were designed to recognise wither the 13bp indel or the wild-type sequence. Sequences were: forward – 5'-GCCAGATTGGACTGGATGTT-3', reverse - 5'-TTATTTCCAGGCCATTCTGG-3', wild-type – 5'-ACGAGTGGAGAATGTGAAGG-3', ed116 – 5'-AAACGGCCCCCTCGTACT-3'. For ed125, the following primers were used to amplify the locus: forward – 5'-TTCAGTTTCAGCTCCTTCCTTC-3', and reverse – 5'-TGGAAGTCAAGTACCAGCA-3'. The PCR product was then digested with the restriction enzyme HpyCH4IV (New England Biolabs, Cat. # R0619S). Additionally, KASP genotyping assays were designed for the ed116 and ed125 alleles (KASP on demand, LGC Biosearch Technologies). Western blots were performed to assess effect of mutations on Foxp1 protein. Protein was isolated from the caudal fin - a pool of seven fin samples was placed on dry ice in RIPA buffer (Thermo, Cat. # 8990) containing protease inhibitor (Thermo, Cat. # A32953). Samples were homogenised with a TissueRupter, centrifuged for 15 mins at 4C, lysate removed and added to LDS blue (Novex, Cat. # B0008) with reducing agent (Novex, B0009), heated to 95C for 10 mins and the centrifuged for 1 min. Protein samples were run on 8% Bis-Tris gels (Invitrogen, NW00082BOX) and run together with protein ladder (Thermo Scientific, Cat. # 26619). Foxp1 antibody was PA5-26848 (Thermo Fisher) and b-actin antibody was (Sigma, A2228). Blood assays were from Biovision (Glucose – K606-100, triacylglyceride – K622-100, cholesterol – K623-100) and performed as per the manufacturer's instructions with 1 ul of blood from an adult zebrafish used for the glucose and cholesterol assays, and 0.2 ul for the triacylglyceride assay. qRT-PCR was performed on a Roche LightCycler 480 using Luna reagents (New England Biolabs,

Cat. # M3003L). cDNA was synthesised using Superscript IV (Invitrogen, Cat. # 18090010). Primers were: *foxp1a* forward 5'-GGCCACTTTGAGGATGACTC-3', *foxp1a* reverse 5'-CCTCGCCACCTAAACTCAG-3', *foxp1b* forward 5'-CATGGCTCCTCTTTACGC-3', *foxp1b* reverse 5'-ACAGCAACGGTAGTGACAGC-3', *bactin* forward 5'-GCCTCCGATCCAGACAGAGT-3' and *bactin* reverse 5'-TGACAGGATGCAGAAGGAGA-3'.

High-fat diet in zebrafish

A high-fat immersion diet was conducted on juvenile fish at 35 dpf. Genotyped fish were stained with Nile Red, imaged on a Leica M205 FCA stereoscope and recovered individually in a 12-well plate whilst the SL of each fish was measured. Based on SL, fish were placed into groups with equal average SL. Within groups, fish were randomly paired up and placed two fish per mesh insert in a 10L tank on the recirculating system, allowing pairs of fish to be tracked through the experiment. The location of each pair was randomly rotated daily to minimise tank location-effects. Fish assigned to the high-fat diet group were immersed in 5% chicken egg yolk for two hours daily over the course of 14 days (16 [↗](#)). Control fish were placed in fresh system water for two hours daily. Both control and high-fat groups received an additional two feeds throughout the day.

Code availability

All code used in this manuscript are available at github.com/jeminchin/morphology_screen [↗](#).

Results

Identification of 102 candidate human adipose morphology genes enriched in adipocyte stem and progenitor cells

To identify candidate regulators of hyperplastic/hypertrophic adipose morphology, we leveraged previously published bulk RNA-Seq data that reported 3,727 differentially expressed genes (DEGs) in human subcutaneous adipose tissue (SAT) characterised by either large or small adipocytes (Fig. 1 [↗](#); Suppl. Table 1 [↗](#)) (27 [↗](#)). We reasoned these DEGs may contribute to establishing hypertrophic or hyperplastic adipose phenotypes. As the RNA-Seq was performed on whole SAT - comprising a heterogeneous mix of cell types - we next asked which adipose-resident cell populations preferentially express these morphology-associated genes. To address this, we clustered the 3,727 candidate genes based on their relative expression across 16 human white adipose tissue (WAT) cell types, using published single-cell transcriptomic data (Fig. 1 [↗](#); Suppl. Fig. 1 [↗](#)) (28 [↗](#)). As expected, the largest subset of morphology DEGs ($n = 730$) was enriched in mature adipocytes, consistent with central roles for adipocytes in defining overall adipose morphology (Suppl. Fig. 1 [↗](#)). Notably, we also identified 102 DEGs enriched in adipose stem and progenitor cells (ASPCs) (Fig. 1 [↗](#); Suppl. Fig. 1; Suppl. Table 2 [↗](#)). Given prior evidence that ASPC abundance and differentiation rate can influence adipose tissue expansion and morphology (36 [↗](#), 37 [↗](#)), we hypothesised that these genes may function within ASPCs to regulate or establish adipose morphology.

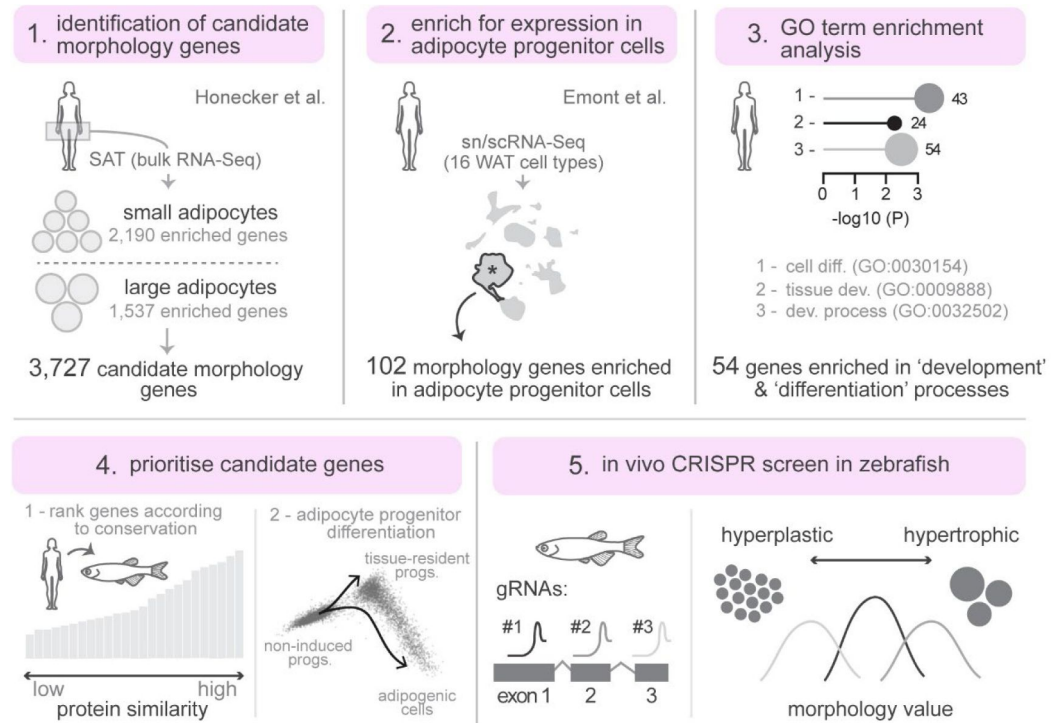


Figure 1.

Project overview.

Depicting steps for candidate gene identification based on human transcriptomic data (1), enriching for genes expressed within adipose stem and progenitor cells, (2) GO term enrichment analysis of candidate genes to focus on development and differentiation processes, (3) prioritisation of candidate genes based on conservation to zebrafish and expression dynamics during adipocyte progenitor differentiation and (4) an in vivo CRISPR screen in zebrafish to functionally characterise genes.

Analysis of candidate human adipose morphology genes using GWAS and cardiometabolic trait associations

Prior to functional analysis, we first explored genetic and cardiometabolic features of the 102 candidate genes. As an initial step, we examined overlap with BMI- and WHR-associated GWAS genes, and found that 16 of the candidate morphology genes were also BMI GWAS genes (*TNRC6B*, *LRMDA*, *PRRX1*, *TSHZ2*, *PTPRD*, *PARD3*, *ADH1B*, *TBX15*, *MAGI2*, *RTL8A*, *CAST*, *BNC2*, *RERE*, *PTPRG*, *ADAMTSL1* and *C1QBP*) and one gene, *HOXC4*, overlapped with WHR-associated loci; however, Fisher's exact test indicated these overlaps were not significantly enriched (Suppl. Fig. 2). We next clustered candidate genes based on their SAT expression correlations with 23 cardiometabolic traits from the METSIM study (29). This revealed two distinct gene-trait signatures: one associated with a "healthy" metabolic profile - characterised by higher hip circumference, fat-free mass, HDL-C, Muscle Insulin Sensitivity Index (MISI) and adiponectin levels, alongside lower BMI, WHR, total fatty acids, triacylglycerides, Insulin and HOMA-B - and another with an "unhealthy" profile, showing the opposite trait pattern (Suppl. Fig. 2). Notably, genes associated with both hypertrophic and hyperplastic adipose morphology were found across both signatures (Suppl. Fig. 2), suggesting that these morphological phenotypes are not strictly aligned with traditional markers of metabolic health.

Enrichment of cell differentiation and developmental genes among adipose morphology candidates

To explore functional themes within our gene set, we performed enrichment analysis to identify overrepresented biological processes. This analysis revealed that 54 of the 102 ASPC-enriched genes were significantly associated with Gene Ontology (GO) terms related to cell differentiation (GO:0030154) and tissue development (GO:0048869, GO:0009888, GO:0032502) (Figs. 1, 2A,B & Suppl. Table 3). Reasoning that they may play key roles in ASPC differentiation and/or the patterning of adipose tissue, we prioritised this subset of 54 genes for further analysis. To refine this list, we applied two complementary strategies. First, we ranked genes by their sequence conservation between humans and zebrafish (Fig. 2D). Second, we assessed scRNA-Seq expression dynamics during progenitor differentiation to adipocytes (38) (Figs. 2D–F). From this, we shortlisted 25 highly conserved genes that were predominantly expressed in either progenitors from undifferentiated samples, or within multipotent progenitors that persist following differentiation (*MGP*⁺ SWAT cells) (Figs. 2D–F & Table 1). This final set included genes with established roles in adipocyte progenitors – such as *PRRX1* (H. Liu et al., 2022), *TBX15* (Pan et al., 2021), and *IRX1* (Han, 2022) – though their specific contributions to adipose morphology have not yet been characterised.

An image-based pipeline to quantify lipid droplet size, number and spatial patterning in zebrafish subcutaneous adipose

To investigate the function of the candidate morphology genes, we developed an image-based phenotyping pipeline in zebrafish. Building on previous studies where Nile Red fluorescence was used to quantify whole-animal adiposity and perform targeted CRISPR screens for regulators of total body fat (16,26), we refined this approach to capture not only global adiposity data but also high-resolution images of Nile Red-positive LDs. This enabled detailed quantification of LD size, number and spatial patterning within zebrafish SAT (Fig. 3A–D). High-magnification images were acquired with a pixel size between 1–2 μm^2 allowing individual LDs to be clearly resolved within zebrafish SAT (Fig. 3A–D). Although very small LDs within multilocular adipocytes fall below our detection threshold, this resolution reliably captures the larger LDs characteristic of mature adipocytes (Fig. 3A–D). Using this pipeline, we characterised SAT development in 107 wild-type zebrafish from three independent clutches (Fig. 1E–G). Representative LD masks illustrating the morphological progression of SAT expansion are shown in Fig. 3E. Consistent with previous observations, LDs are first deposited at the anterior end of the adipose depot and new LDs are added in both posterior and dorsoventral directions (Fig. 3E). To spatially track this progression, we annotated the most anterior LD and defined successive 200 μm strata extending posteriorly from this reference point (Fig. 3E). This stratification enabled comparisons of LD morphology and number at anatomically and developmentally equivalent positions across individuals. As zebrafish grew from 7 to 12 mm standard length (SL), we observed limited increases in LD number within the anterior-most stratum (stratum 1) (Fig. 3E & F). Instead, new LDs accumulated in more posterior regions, particularly in strata 3 and 4, which exhibited the greatest LD number and overall growth (Figs. 3E & F). Concurrently, LDs increased in size, reaching an average diameter of $\sim 65 \mu\text{m}$ by 11–12 mm SL (Fig. 3E & G). Notably, posterior strata – containing more recently formed LDs – appeared to be in the process of enlarging toward this consistent size benchmark (Fig. 3E & G). Together, these findings show that during these relatively young juvenile stages, LDs expand rapidly in size but ultimately stabilise at an average diameter of $\sim 65 \mu\text{m}$, reflecting a potential upper limit of LD growth at this developmental stage.

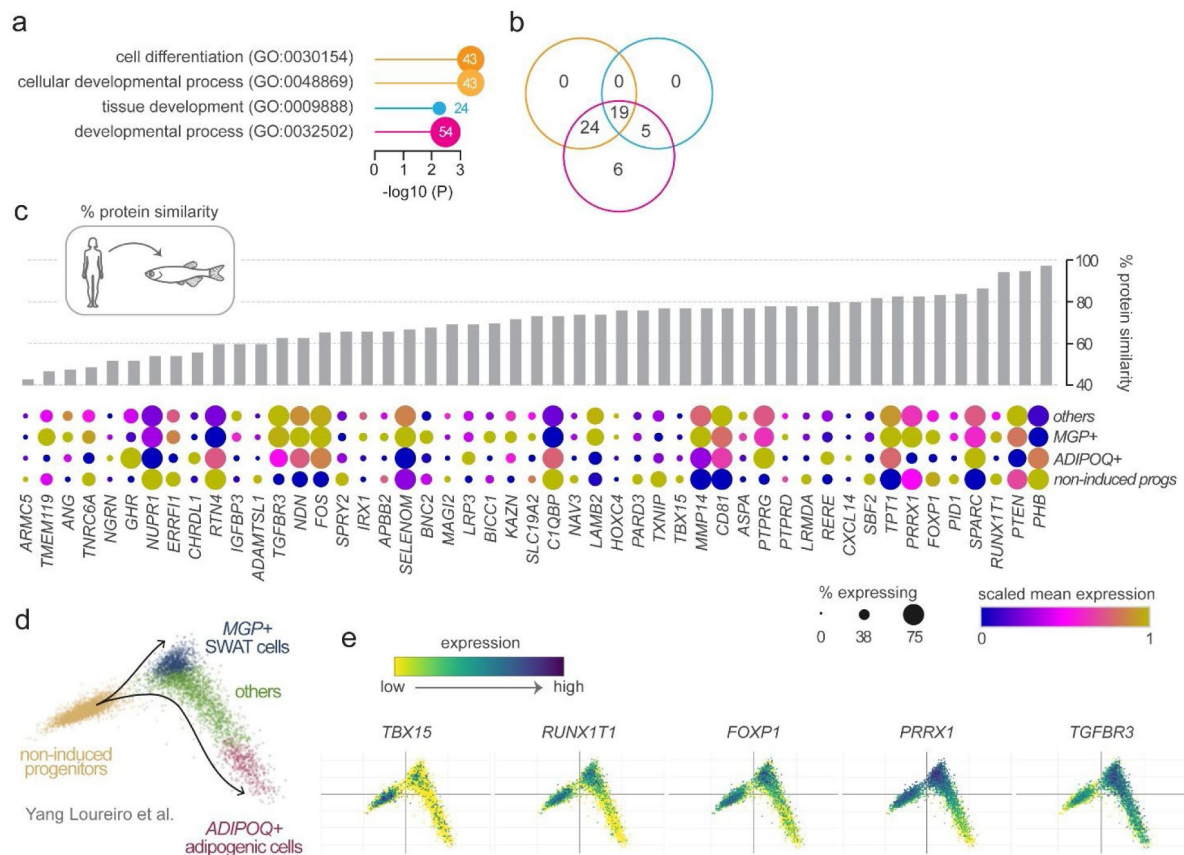


Figure 2.

Prioritisation of candidate adipose morphology genes based on conservation to zebrafish and expression dynamics during progenitor differentiation.

A. GO term enrichment analysis of the 102 candidate morphology genes enriched in adipose stem and progenitor cells. 54 of these genes were enriched in four GO terms related to development and differentiation. **B.** Venn diagram showing gene intersection between GO terms. Cell differentiation (GO:0030154) and cellular developmental process (GO:0048869) had 100% overlap. **C.** Grey bars denote % protein similarity between human and zebrafish orthologs. Dot plots below show scaled mean expression (colour) and % of cells expressing (dot size) each candidate genes within four cell types identified during progenitor differentiation to adipocytes. Cell types are: non-induced progenitors, MGP+ SWAT cells, ADIPOQ+ adipogenic cells and others, based on Yang Loureiro et al., 2023. **D.** PCA projection of single cells according to cluster designation from C. **E.** Example candidate morphology genes and their expression during adipocyte differentiation.

Table 1.

Zebrafish gene targets and adipose morphology statistics.

Adjusted p values after Benjamini-Hochberg FDR correction for 23 statistical tests. Small, medium and large correspond to effect sizes (Cohen's d). KS = Kolmogorov-Smirnov test. LMM = linear mixed model with experiment as random effect. Values in bold indicate statistical significance. All genes were targeted in two independent replicates (except for *aspa* and *lamb2* which only had a single replicate).

gene target (zebrafish)	sample size		Statistical power			KS test (FDR adjusted p value)	LMM – slope (FDR adjusted p value)	LMM – intercept (FDR adjusted p value)
	expt.	cont.	Small (0.4)	Medium (0.6)	Large (1)			
<i>prrx1b</i>	51	57	54%	87%	99.9%	0.2652	0.979	0.871
<i>foxp1b</i>	36	57	46%	80%	99.6%	0.02176	0.0022632	0.021712
<i>runx1t1</i>	22	34	30%	58%	95%	0.4287	0.232961	0.306659
<i>ptenb</i>	25	29	30%	58%	95%	0.006846	0.674	0.508
<i>cacnb4a</i>	23	34	31%	59%	95%	0.6418	0.288296	0.463958
<i>sparc</i>	23	35	31%	59%	95.6%	0.293	0.557	0.767
<i>pid1</i>	26	29	31%	59%	95.3%	0.3481	0.10398	0.15810
<i>sdk1a</i>	25	34	32%	61%	96%	0.7517	0.534960	0.670365
<i>cxcl14</i>	24	29	30%	57%	94%	0.02255	0.462406	0.682372
<i>ptprga</i>	35	39	40%	72%	99%	0.4389	0.769939	0.810014
<i>ptprdb</i>	24	39	33%	62%	97%	0.203	0.212	0.450
<i>lrmda</i>	22	29	28%	55%	93%	0.966	0.181489	0.206508
<i>txnipa</i>	24	36	32%	61%	96%	0.03319	3.10e-05	0.000923
<i>tmem115</i>	29	29	32%	61%	96%	0.1469	0.224	0.187
<i>tbx15</i>	17	35	26%	51%	91%	0.3098	0.1416	0.1199
<i>mmp14b</i>	30	36	36%	67%	98%	0.03856	0.00737	0.07725
<i>cd81a</i>	26	35	33%	63%	97%	0.5928	0.160	0.198
<i>aspa</i>	17	17	20%	40%	81%	0.778		
<i>srpx</i>	20	25	26%	50%	90%	0.02411	0.263	0.201
<i>pard3ab</i>	32	42	39%	71%	99%	0.7033	0.545	0.592
<i>nav3</i>	28	42	37%	68%	98%	0.3718	0.199	0.258
<i>lamb2</i>	6	21	13%	24%	55%	0.7665		
<i>kazna</i>	23	42	33%	62%	97%	0.293	0.0774	0.0958

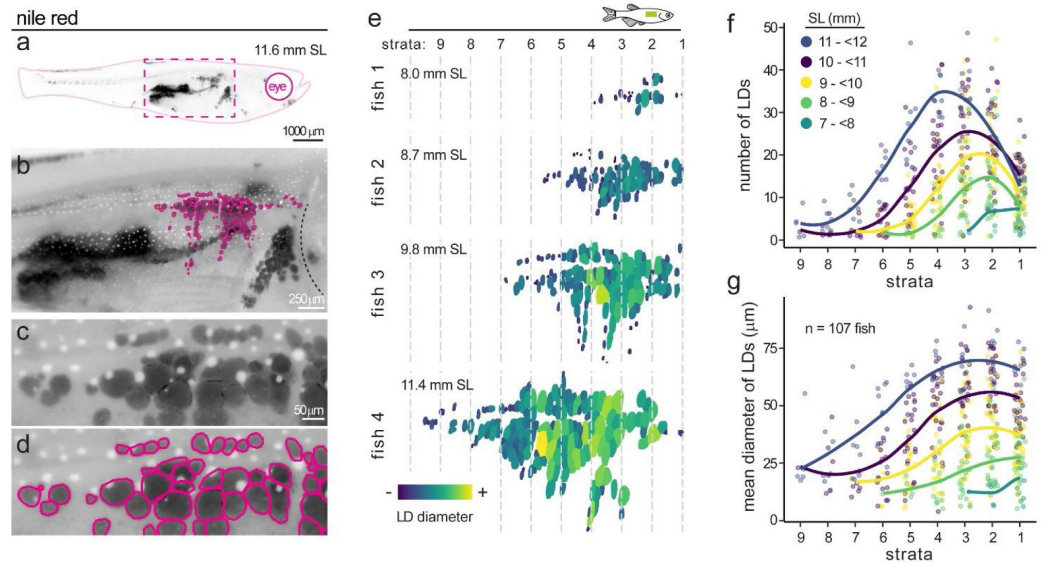


Figure 3.

Spatial dynamics of subcutaneous adipose growth in zebrafish.

A. Nile Red stained zebrafish at 11.6 mm standard length (SL). Black signal denotes Nile Red+ neutral lipid within adipose. The magenta dotted box denotes the area shown in **B**. **B.** Higher magnification image showing region of interest. Magenta outlined lipid droplets (LDs) highlight the subcutaneous adipose tissue. **C.** Zoomed in view of subcutaneous adipose LDs. White dots are melanosome pigment granules. **D.** Magenta lines outline the LD segmentation performed on LDs from **C**. **E.** Segmented subcutaneous adipose LDs colour-coded according to LD diameter from four representative zebrafish at different sizes. Fish sizes are shown in mm SL. Strata were defined in 200 μ m intervals from the most anterior (to the right) LD. **F.** Number of LDs in each strata according to size of the zebrafish (SL). Zebrafish were categorised into 5 groups according to size. **G.** Mean diameter of LDs in each strata according to size of zebrafish.

Quantification of adipose morphology in juvenile zebrafish

As LD size, number and spatial dynamics can be robustly captured in juvenile zebrafish, we next developed methodology to quantify adipose morphology – specifically hyperplastic versus hypertrophic patterns – in a format scalable for CRISPR-based screening. Our approach was conceptually based on previously described methods for assessing human adipose morphology (35). To establish a baseline, we quantified SAT LD number and size within 194 wild-type fish (including the 107 animals analysed in **Fig. 3**). We observed a strong curvilinear relationship between mean LD diameter and the total depot size ($R^2 = 0.85$) (**Fig. 4A**). This relationship also held true for other LD size metrics (including area and perimeter), further suggesting that as the zebrafish SAT expands, LDs initially grow rapidly in size but then plateau as they reach a steady-state size and hypertrophic growth slows relative to the overall size of the depot. Following the strategy of Arner et al (2010), we calculated a ‘morphology value’ for each individual based on the deviation of each fish from the fitted curve (**Fig. 4B**). Morphology values were normally distributed around zero, with approximately equal proportions of individuals exhibiting hyperplastic (negative deviation) or hypertrophic (positive deviation) SAT (**Fig. 4B**). Notably, morphology values were inversely correlated with LD number ($p = 3.9 \times 10^{-6}$), such that

hypertrophic individuals had fewer, larger LDs while hyperplastic individuals had more numerous, smaller LDs (**Fig. 4C**). Together, these data show that morphology values can be applied to zebrafish SAT and used to quantify hyperplastic/hypertrophic patterning.

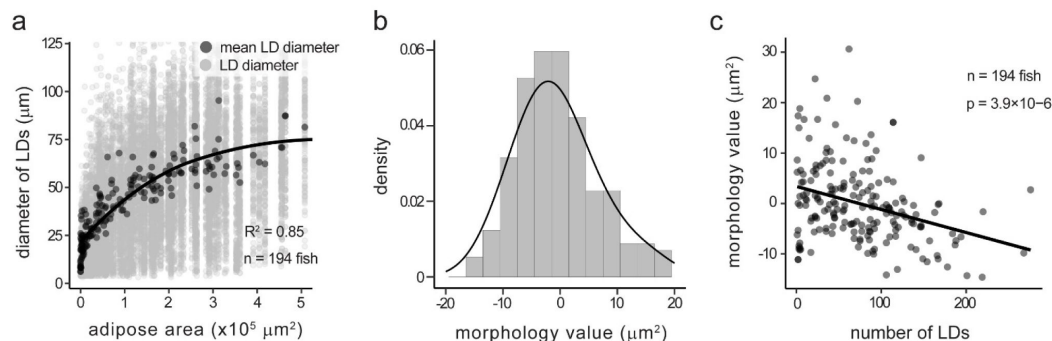


Figure 4.

Calculation of adipose morphology value based on relationship between lipid droplet size and overall adipose depot size.

A. Diameter of subcutaneous adipose LDs plotted relative to depot area. Mean LD diameters per fish are dark grey, individual LD diameters are light grey. Line was fitted using a generalised additive model. **B.** Morphology values were calculated as the deviation of mean LD diameter from the fitted model in A. Note the normal distribution of morphology values around zero. **C.** Inverse relationship between morphology value and number of LDs. A linear model was fitted to show the inverse relationship.

Morphology values can be used to assess differences in F0 zebrafish CRISPR mutants

We next tested whether this metric could detect known regulators of adipose morphology. As a positive control, we targeted *growth hormone 1* (*gh1*) using established F0 CRISPR mutant methodology. Consistent with previous findings that *gh1* mutants display hypertrophic SAT LDs (39), *gh1* F0 CRISPR mutants displayed significantly larger LDs and an upward shift in morphology values ($p = 0.04$) (Suppl. Fig. 3). Notably, this effect was detected in only nine *gh1*-targeted fish, corresponding to a large effect size (Cohen's $d = 1.06$) (Suppl. Fig. 3). Power analysis based on this dataset indicated that similar effects could be detected with 80% power using as few as 11 fish. Moreover, detecting medium-sized effects (Cohen's $d = 0.6$ - 0.7) would require only 23-31 mutant fish. Together, these results demonstrate that image-based profiling of adipose morphology is a sensitive and scalable approach for in vivo genetic screens aimed at identifying regulators of hyperplastic and hypertrophic adipose growth.

A targeted CRISPR screen identified six genes that alter subcutaneous adipose morphology in zebrafish

Following successful validation of our approach by targeting of *gh1*, we applied the same F0 CRISPR methodology and image-based pipeline to functionally assess all 25 conserved candidate genes identified from human adipose tissue datasets. Two independent experimental replicates were performed for each gene target (Table 1), resulting in a total of 50 CRISPR targeting experiments and 1,393 juvenile zebrafish screened (Table 1). To quantify SAT morphology and classify hyperplastic or hypertrophic profiles, we applied two complementary statistical approaches. First, we calculated morphology values for each fish and compared mutant versus

control morphology value distributions using the Kolmogorov-Smirnov (KS) test. This non-parametric test does not assume a specific distribution and is sensitive to shape-based distributional changes. Second, we applied linear mixed-models (LMMs) to evaluate the relationship between LD size and total SAT area, whilst accounting for replicate-specific variability. The LMMs allowed us to test whether LDs scaled normally with tissue size (slope) or differed in size at any given adipose area (intercept). After correcting for multiple tests, the KS test identified three mutants with hypertrophic morphology (*foxp1b*, *txnipa* and *mmp14b*) and three with hyperplastic morphology (*ptenb*, *cxcl14* and *srpx*) (Table 1 & Fig. 5). Among the hypertrophic mutants, *foxp1b* and *txnipa* were also detected using both LMMs for slope and intercept, whilst *mmp14b* was only identified using the LMM for slope (Table 1 & Fig. 5). Conversely, hyperplastic morphology in *ptenb*, *cxcl14* and *srpx* mutants was only detected using the KS test (Table 1 & Fig. 5). Interestingly, *ptenb* mutants exhibited both hyperplastic and hypertrophic populations of LDs, suggesting a multimodal distribution of LDs (Fig. 5A). Together, these findings demonstrate image-based profiling is an effective and scalable approach for identifying regulators of adipose morphology.

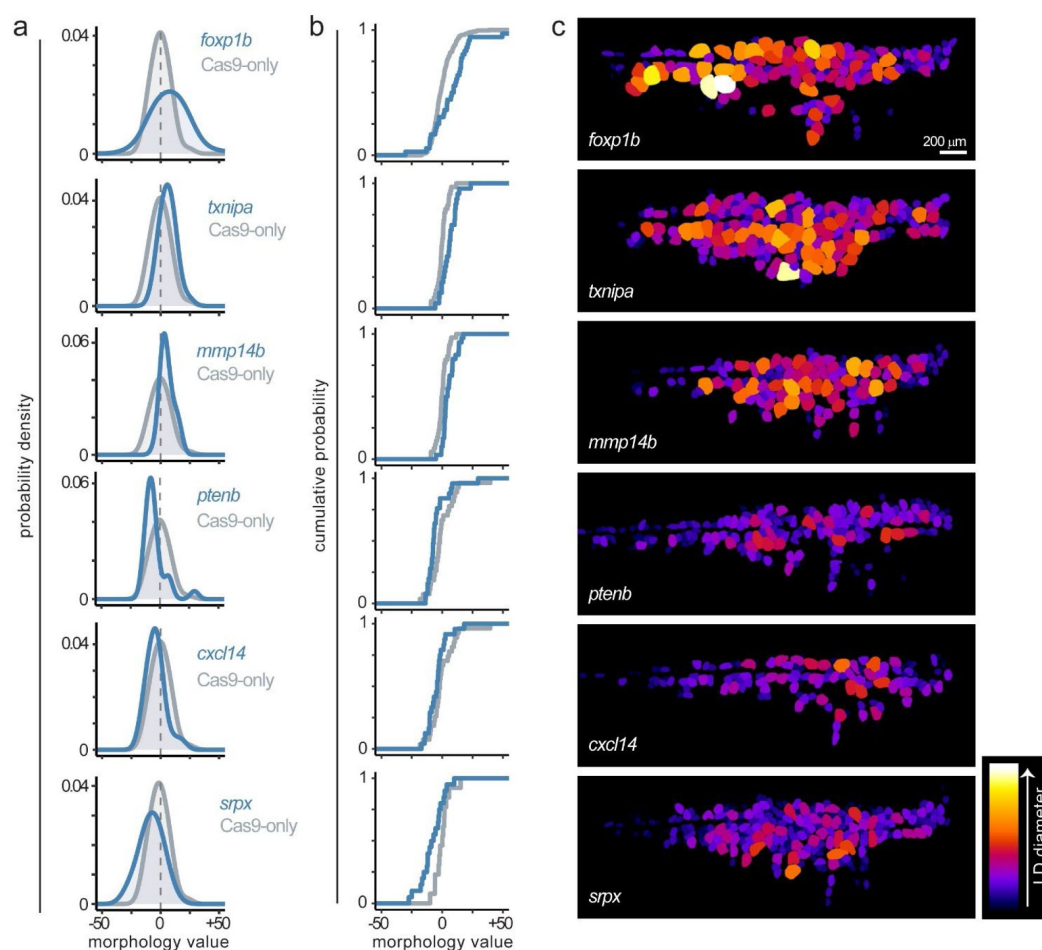


Figure 5.

Identification of mutants with hypertrophic or hyperplastic adipose morphology.

A. Morphology values for individual mutants (blue) with respect to their Cas-9 only control (grey). Dotted vertical line represents a morphology value of zero. **B.** Cumulative probability functions showing shift towards hypertrophic (*foxp1b*, *txnipa* and *mmp14b*) or hyperplastic (*ptenb*, *cxcl14* and *srpx*) morphology. **C.** Segmented subcutaneous adipose LDs from each respective mutant colour-coded according to LD diameter.

A hypertrophic growth bias of subcutaneous adipose in *foxp1b* and double *foxp1a;foxp1b* zebrafish mutants

Notably, targeting *foxp1b* (one of two zebrafish Foxp1 genes) induced hypertrophic SAT morphology (Fig. 5). Foxp1 is required for maintaining undifferentiated and self-renewing stem and progenitor cells in multiple tissue compartments (40–44). Further, in mouse, Foxp1 inhibits brown adipogenesis and browning of white adipocytes (45). However, roles for Foxp1 in establishing hyperplastic/hypertrophic morphology of adipose is unknown. To further study the role of Foxp1, we generated new loss-of-function alleles for both *foxp1* genes (*foxp1a* and *foxp1b*) in zebrafish (Fig. 6A). We targeted the highly conserved Forkhead domain (FHD) (Fig. 6B), which is essential for DNA binding and therefore Foxp1 transcriptional activity (46,47) (Fig. 6B). We generated two alleles – *foxp1a^{ed116}* and *foxp1b^{ed125}* – which each contained a frameshift and premature stop codon within the FHD (Fig. 6B). qRT-PCR revealed reduced *foxp1a* or *foxp1b* mRNA in the respective single mutants, with no apparent mRNA upregulation of the paralogous *foxp1* gene, or other members of the Foxp family (Suppl. Fig. 4). Western blots revealed a significant reduction of Foxp1 protein in each of the single mutants, and in *foxp1a;foxp1b* double mutants suggesting severe loss-of-function alleles (Fig. 6C & Suppl. Fig. 4). At approximately one-month of age (~10 mm SL), *foxp1b^{ed125}* and double *foxp1a;foxp1b* mutants were significantly smaller than either wild-type control or *foxp1a^{ed116}* fish (Fig. 6E), and analysis of baseline lipid storage within adipose tissues using Nile Red revealed that *foxp1b^{ed125}* and double *foxp1a;foxp1b* mutants also had lower lipid storage in adipose tissues than control fish (normalised to size of fish) (Fig. 6F). Similar to that observed in F0 CRISPR mutants, at one-month of age *foxp1b^{ed125}* and *foxp1a;foxp1b* mutants had significantly larger LDs characteristic of hypertrophic morphology (Fig. 6G & H). Analysis of blood metabolites revealed *foxp1a;foxp1b* double mutants were hyperglycaemic (Suppl. Fig. 5), and strikingly, in *foxp1a;foxp1b* double mutants there was increased lipid accumulation in the liver (Fig. 6D & Suppl. Fig. 5D). At adulthood, fish size and lipid storage within adipose in *foxp1a;foxp1b* double mutants had largely caught up with wild-type control fish (adiposity at ~80% of wild-type levels) (Suppl. Fig. 6). These findings show that stable *foxp1b* mutants have hypertrophic adipose along with metabolic dysfunction.

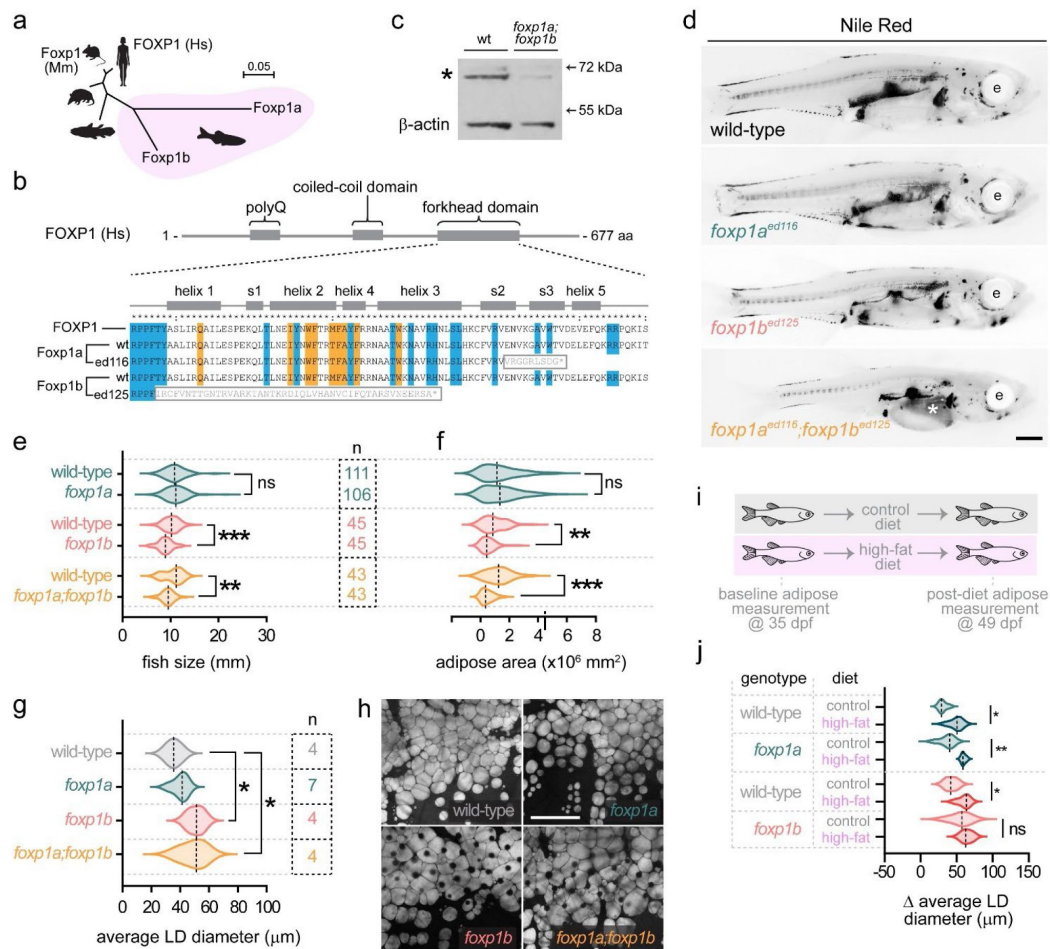


Figure 6.

Stable *foxp1b* zebrafish mutants have hypertrophic adipose but are unable to undergo hypertrophic expansion in response to a high-fat diet.

A. Phylogenetic tree showing relatedness of zebrafish Foxp1a and Foxp1b amino acid sequences to human, mouse, opossum and ceolocanth Foxp1. Bar indicates substitutions per X. **B.** Overview of human Foxp1 domain structure showing a polyQ, coiled-coil and forkhead domain. Zoomed view of the DNA-binding forkhead domain showing structural features including helix and S. Essential amino acids of DNA binding are highlighted in orange, amino acids essential for DNA binding are highlighted in blue. Zebrafish wild-type Foxp1a and Foxp1b sequences are shown aligned to human, along with the ed116 foxp1a and ed125 foxp1b mutant alleles. Grey boxes show the addition of nonsense peptide followed by premature stop codon. **C.** Western blot showing the reduction of Foxp1 protein in double *foxp1a;foxp1b* zebrafish mutants. B-actin is used as a loading control. Asterisk indicates the Foxp1 protein band. **D.** Nile Red fluorescence images showing adipose lipid distribution (black signal) in wild-type, *foxp1a*^{ed116}, *foxp1b*^{ed125} and double *foxp1a*^{ed116}; *foxp1b*^{ed125} zebrafish mutants. E indicates the eye. Asterisk indicates lipid accumulation within the liver in double mutants. Scale bar is 1 mm. **E.** Violin plots showing fish size (standard length, mm) of *foxp1a*^{ed116}, *foxp1b*^{ed125} and double *foxp1a*^{ed116}; *foxp1b*^{ed125} mutants compared to wild-type siblings. **F.** Violin plots showing Nile Red-positive adipose area in *foxp1a*^{ed116}, *foxp1b*^{ed125} and double *foxp1a*^{ed116}; *foxp1b*^{ed125} mutants compared to wild-type siblings. **G.** Violin plots showing average LD diameter in *foxp1a*^{ed116}, *foxp1b*^{ed125} and double *foxp1a*^{ed116}; *foxp1b*^{ed125} mutants compared to wild-type siblings. **H.** LipidTox stained LDs within subcutaneous adipose of *foxp1a*^{ed116}, *foxp1b*^{ed125}, double *foxp1a*^{ed116}; *foxp1b*^{ed125} mutants and wild-type sibling. **I.** Overview of high-fat diet feeding experiment where zebrafish were genotyped and Nile Red imaged at 35 dpf, before undergoing a 14-day high-fat diet immersion (5% chicken egg yolk) for two hours daily in addition to normal feeding regimen.

Control diet was two hour immersion in system water daily. Post-diet Nile Red imaging was performed at 49 dpf. J. Violin plots showing effects of diet and genotype on average LD diameter. Statistical tests were one-way ANOVA followed by a Tukey's HSD post-hoc test.

***foxp1b* zebrafish mutants are unable to undergo hypertrophic remodelling in response to a high-fat diet**

Our previous analyses showed that *foxp1b* mutants display a bias toward hypertrophic growth during developmental expansion of SAT. To determine whether this bias impacts the adipose response to dietary challenge, we subjected *foxp1a*, *foxp1b*, and *foxp1a;foxp1b* double mutants to a two-week high-fat diet (HFD), a condition previously shown to induce SAT expansion (16). All *foxp1* mutant genotypes exhibited reduced overall adipose expansion compared to control fish, indicating a general impairment in expansion capacity (Suppl. Fig. 7). However, when LD diameter was assessed, a striking difference emerged: the already hypertrophic SAT in *foxp1b* mutants failed to undergo further hypertrophic remodelling in response to HFD (Fig. 6J). In contrast, *foxp1a* mutants, which did not display a hypertrophic phenotype at baseline, demonstrated clear hypertrophic expansion following HFD exposure (Fig. 6J). These findings suggest that the hypertrophic bias observed in *foxp1b* mutants during development does not confer an enhanced capacity for hypertrophic remodelling during dietary-induced adipose growth. Taken together, these results indicate that stable *foxp1b* and *foxp1a;foxp1b* double mutants exhibit hypertrophic adipose morphology, but are unable to further remodel hypertrophically in response to nutritional excess.

Discussion

In this study, we (i) identified candidate human genes implicated in hypertrophic and hyperplastic remodelling of adipose tissue, (ii) established an image-based profiling method for assessing adipose morphology in zebrafish, (iii) conducted an in vivo CRISPR screen in zebrafish to functionally evaluate the roles of 25 candidate genes, (iv) identified six genes that influence hyperplastic or hypertrophic remodelling, (v) focused on *Foxp1*, generating stable zebrafish mutant lines that recapitulate phenotypes observed in F0 CRISPR experiments, and (vi) discovered that although *foxp1b* mutants display baseline hypertrophic morphology, they fail to undergo further hypertrophic remodelling in response to a high-fat diet, suggesting that a developmental bias in adipose growth influences later responses to diet. Together, this work demonstrates a scalable, tractable screening platform to identify regulators of adipose remodelling.

In this study we employed a multi-step process to identify candidate morphology genes from human data. As GWAS for adipose morphology has only been conducted in relatively small cohorts (48,49), we instead used published transcriptomic data that identified DEGs enriched in SAT characterised by large or small adipocytes (27). Clustering of the morphology DEG expression correlations with cardiometabolic traits separated genes into 'healthy' and 'unhealthy' profiles (Suppl. Fig. 2); genes associated with larger adipocytes generally aligned with adverse metabolic traits such as increased BMI and WHR in accordance with adipocyte size increasing with BMI (50). However, this pattern was not absolute—some candidates diverged from this axis (Suppl. Fig. 2), representing potentially interesting regulators of adipose morphology that are uncoupled from traditional metabolic or disease states. For this study, we focussed on candidate genes that were highly expressed or enriched within adipocyte stem and progenitor cells (ASPCs), wanting to focus on the hyperplastic potential of adipose influencing remodelling capacity and mechanism. However, we note that hyperplastic adipose potential is likely determined by multiple cell types, not just progenitors, and also including the adipose microenvironment (51,52).

We established a rapid, scalable, stereomicroscope-based imaging pipeline for profiling zebrafish adipose morphology. Our imaging resolution was sufficient to detect large LDs but not smaller LDs found as secondary LDs in zebrafish adipocytes (52 [↗](#)), a phenomenon which is also found in mouse adipocytes (53 [↗](#)). Nevertheless, our methodology reliably captured large LDs and allowed us to use LD size as a proxy for overall adipocyte size. We adapted an existing analytical framework from the Arner lab, applying morphology value metrics developed for human adipose tissue to zebrafish SAT, and observed very similar relationships, including a curvilinear relationship between LD size and depot size and an inverse correlation between LD number and LD size. We therefore believe that the growth dynamics of zebrafish adipose is likely similar to that of mammals. To test for differences in adipose morphology, we implemented two complementary statistical approaches: the Kolmogorov–Smirnov (KS) test to compare the overall distribution of morphology values between groups, and linear mixed models (LMMs) to assess group-specific changes in LD size while accounting for experimental batch effects. The KS test proved more sensitive to subtle distributional shifts, while the LMMs provided a more conservative measure of structured group-level differences. This dual-method strategy is well-suited for future high-throughput screens, including those targeting GWAS-derived candidates.

Using this platform, we identified six mutants that altered adipose morphology. *Txnipa* (*txnipa*) F0 CRISPR mutants displayed hypertrophic morphology. TXNIP (Thioredoxin-Interacting Protein) is a well-characterised regulator of oxidative stress, metabolism, and inflammation. TXNIP binds and inhibits the antioxidant thioredoxin, leading to increased production of reactive oxygen species and oxidative stress (54 [↗](#)). Interestingly TXNIP links nutrient sensing to inflammatory signalling by responding to and regulating glucose uptake and NLRP3 inflammasome complex formation (54 [↗](#)). Interestingly, the ROS/TXNIP/NLRP3 pathway impairs insulin signalling, a major driver of adipogenesis and hyperplastic adipose growth (55 [↗](#)). Further, vitamin D increases TXNIP expression (56 [↗](#)), which plays a major role in energy storage in zebrafish (57 [↗](#)). *Mmp14b* F0 zebrafish mutants also exhibited hypertrophic adipose. MMP14 (Matrix Metalloproteinase 14) is a pericellular collagenase and plays a key role in ECM remodelling. In mice, MMP14 deficiency stiffens the adipose matrix and restricts hypertrophic expansion, resulting in adipose characterised by small adipocytes (58 [↗](#)). Conversely, overexpression of *Mmp14* increased the size of adipocytes (59 [↗](#)). As such, these mouse phenotypes appear contradictory to the hypertrophic morphology we observe after mutation of *mmp14b* in zebrafish and warrant further investigation. We note that two *mmp14* genes exist in zebrafish, therefore both paralogues could be targeted in future studies. *Ptenb* F0 zebrafish mutants predominantly showed a hyperplastic phenotype, however a population of hypertrophic LDs were also present suggesting a bimodal size distribution. PTEN (Phosphatase and Tensin homolog) inhibits PI3K/AKT signalling and loss in progenitor cells promotes differentiation and hyperplastic growth in line with formation of lipomas in patients with PTEN hamartoma tumour syndrome (60 [↗](#)). Interestingly, PTEN knockout mouse models also appear to show bimodal adipocyte sizes in gonadal and retroperitoneal adipose, suggesting that complex adipose expansion dynamics are controlled by PTEN (61 [↗](#)). *Cxcl14* F0 zebrafish mutants also displayed a hyperplastic phenotype. CXCL14 is a chemokine and adipokine that regulates immune tone and progenitor dynamics in adipose tissue. Circulating *CXCL14* mRNA was decreased in obesity, and inversely correlated with pro-inflammatory pathway members, and positively correlated with GLUT4 and Adiponectin suggesting a dysregulated metabolic and inflammatory state (62 [↗](#)).

Of great interest, *srpx* F0 CRISPR mutants showed a hyperplastic SAT morphology. Interestingly, SRPX has not been previously shown to have a role in adipose tissue, however it is part of the transcriptomic signature of CD142+ ASCs in mouse, which function as non-adipogenic inhibitors of adipogenesis (Aregs) (63 [↗](#)). It is therefore of interest to test whether *Srpx* functions within Aregs to regulate hyperplastic remodelling, with a reduction of *Srpx* leading to increased hyperplastic growth. Outside of adipose, *Srpx* is implicated in the negative regulation of cell proliferation and has a marker gene for glioblastoma (64 [↗](#)).

For the primary functional follow-up in this study, we focused on *Foxp1*, a gene known to maintain stem and progenitor populations in multiple tissues including the hair follicle, haematopoietic niche, mammary gland, and cerebral cortex (40 [40](#), 41 [41](#), 43 [43](#), 44 [44](#)). We hypothesise that *Foxp1* may be required in adipose progenitors for hyperplastic growth, and that its loss would result in compensatory hypertrophy. In mouse models, *Foxp1* deletion leads to increased browning of white adipose tissue and enhanced thermogenesis (45 [45](#)), suggesting a repressive role in thermogenic programming. Although zebrafish are ectothermic and historically not thought to possess thermogenic adipocytes, recent evidence shows epicardial adipose in zebrafish is thermogenic (15 [15](#)), raising the possibility that *foxp1* mutants may also undergo browning—an intriguing hypothesis for future work. We generated stable *foxp1a* and *foxp1b* mutant lines carrying severe loss-of-function alleles, confirmed by reduced protein expression. *Foxp1b* mutants displayed more severe phenotypes than *foxp1a*, consistent with its higher conservation with mammalian *FOXP1*. The major phenotype in *foxp1a* mutants was impaired adipose expansion following HFD, suggesting failure to respond to diet-induced stress signals. In contrast, *foxp1b* mutants were smaller at baseline, with less adipose tissue, along with a failure to expand adipose following a HFD. Notably, the double mutants developed fatty liver and hyperglycaemia—phenotypes also observed in *Foxp1*-deficient mice (65 [65](#)), where increased hepatic de novo lipogenesis (DNL) contributes to ectopic fat deposition and disrupted glucose homeostasis. This may similarly underlie the zebrafish phenotype and could be tested in future work.

Strikingly, despite *foxp1b* mutants displaying a baseline bias towards hypertrophy, they were unable to undergo further hypertrophic expansion in response to HFD. This contrasts with *foxp1a* mutants, which readily increased LD size post-diet. The inability of *foxp1b* mutants to undergo diet-induced hypertrophic remodelling suggests a potential ceiling effect on adipocyte size, possibly due to physical or molecular constraints. As all experiments were conducted in growing juveniles, this may reflect developmental limitations—such as restricted compartment space or altered ECM properties. Alternatively, *foxp1b* mutants may fail to perceive or respond to HFD-induced molecular cues, potentially implicating *Foxp1* in metabolic sensing (66 [66](#)). It remains to be determined whether this is a *foxp1b*-specific effect or if a general developmental bias towards hypertrophy limits later hypertrophic remodelling which could be tested in the other hypertrophic mutants identified in this screen.

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Additional information

Author contributions

RW and PT performed experiments, RW, PT and JENM performed data analysis, PT and JENM wrote the manuscript.

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Additional files

Supplemental Table 1 [↗](#)

Supplemental Table 2 [↗](#)

Supplemental Table 3 [↗](#)

Supplemental Table 4 [↗](#)

Supplementary Figures [↗](#)

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Joint Public Review:

In this manuscript, Wafer and Tandon et al. present a thoughtful and well-designed genetic screen for regulators of adipose remodeling using zebrafish as a model system. The authors cross-referenced several human adipocyte-related transcriptomic and genetic association datasets to identify candidate genes, which they then tested in zebrafish. Importantly, the authors devised an unbiased microscopy-based screening platform to document quantitative adipose phenotypes with whole animal imaging, while also employing rigorous statistical methods. From their screen, the authors identified 6 genes that resulted in robust adipose phenotypes out of a total of 25 that were tested. Overall, this work will be a useful resource for the field because of both the genes identified and the quantitative, rigorous screening pipeline. However, there are limitations that preclude a definitive distinction between developmental and remodeling effects that should be acknowledged and discussed, or addressed with new experiments.

Strengths:

- (1) This work combines multiple omic datasets to identify candidate genes that informed a CRISPR-based screen to identify genes underlying adipose tissue development and adaptation. This approach offers a new avenue to improve our understanding and testing of new genetic mechanisms underlying the development of obesity.
- (2) Using a clever screening approach, this study identifies new genes that are associated with adipose tissue lipid droplet size change. Importantly, the study provides further validation using a stable CRISPR line to show the phenotype in basal and high-fat diet conditions.
- (3) The experiments are well-designed and rigorous. Sample sizes are large. Statistical analyses are highly rigorous, contributing to a high-quality study.

Weaknesses:

(1) The image quantification established in Figures 3 and 4 and used in CRISPR screening showed the relationship among zebrafish development, adipose tissue size, and lipid droplet size. Although adipose tissue development patterning is linked with adipose tissue adaptation, as shown by the evidence provided in this paper, it will be more powerful if the imaging method and pipeline were established to directly access the adipose tissue plasticity rather than just the developmental patterning. Furthermore, the authors should perform additional analysis of their existing data to more accurately determine lipid droplet size along the AP axis in response to HFD.

(2) In the absence of tissue-specific manipulations, definitively establishing the mechanisms underlying the genetic regulation of adipose tissue physiology presents limitations.

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